Devanshu Kumar Ranjan

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Summary

AI Engineer at KAS Global Commerce and M.Tech Cyber Security graduate from NIT Patna (CGPA: 9.06/10), specializing in production AI systems, LLM agent workflows, and privacy-preserving machine learning. Architect of the enterprise Root Cause Analysis (RCA) Diagnostic Engine: a LangGraph multi-agent investigator explaining complex advertising performance anomalies in natural language across 15 production diagnostics and 50+ deterministic rules. Engineered database optimizations slashing PostgreSQL analytics query latency by 88%+ (from >1 minute to <7 seconds) via multi-query parallelization and compound indexing. Scaled automated catalog harvesting, normalization, and multimodal vision intelligence (Gemini 2.5 Flash Vision) to 9 global marketplaces with zero manual intervention. Master's thesis on locally differentially private (LDP) federated sequential recommendation. Experienced in designing resilient, low-latency AI backends with Python, FastAPI, PyTorch, LangGraph, and PostgreSQL with `pgvector`.

Education

National Institute of Technology, Patna — Patna, India
M.Tech., Cyber Security (CSE) — CGPA: 9.06/10 — *August 2024 – May 2026*

Graphic Era University, Dehradun — Dehradun, India
B.Tech., Computer Science and Engineering — CGPA: 8.83/10 — *June 2019 – June 2023*

Sri Chaitanya Techno School (CBSE) — Visakhapatnam, India
Senior Secondary — 80.6% — *2019*

Cambridge School (ICSE) — Cuttack, Odisha
Secondary — 90.17% — *2017*

Experience

KAS Global Commerce Inc. — Bilaspur, HP (On-site)
AI Engineer — *July 2026 – Present*

- Architected and deployed a production LangGraph multi-agent RCA engine from scratch (research to deployment): **15 live diagnostics and 50+ root-cause rules** turning seller queries (English/Hinglish) into automated anomaly investigations; engineered input sanitization and security guardrails to prevent prompt injection; reduced PostgreSQL analytics fetch latency from >**1 minute down to <7 seconds (88%+ speedup)** via multi-query parallelization, compound B-tree/GIN indexing, and optimized relational joins.
- Cut monthly catalog harvesting and multimodal vision processing overhead to **zero manual intervention** via scheduled cron orchestrators and automated schema recovery.
- Scaled catalog harvesting, normalization, and vision intelligence to **9 global marketplaces** (Amazon.in, Amazon.com, ONDC, Walmart, Flipkart, eBay, etc.).
- Analyzed visual design patterns, product placement, background settings, and headline messaging across competitor Amazon A+ content to generate structured insight JSONs for automated A+ content creation.

KAS Global Commerce Inc. — Bilaspur, HP (On-site)
AI Research Intern — *September 2025 – June 2026*

- **Distributed Two-Stage Web Scraping Engine:** Engineered high-volume catalog harvesting across **Amazon.in**, **Amazon.com**, and **ONDC**; reduced full-catalog scraping time from **2–3 months down to 1.5 weeks** (90%+ latency reduction) via threaded Zyte API calls at **1,000 req/min** with MongoDB persistence.
- **Taxonomy ETL & Dense Vector Search:** Developed taxonomy breadcrumb normalization and dense semantic vector search using OpenAI `text-embedding-3-large` (3,072 dimensions) and PostgreSQL with `pgvector` (IVFFLAT indexing) for cross-marketplace category mapping, achieving **sub-50ms retrieval latency**.
- **Common API Endpoints & Chatbot RAG:** Built 3 core microservices (`/api/v1/category-mapper`, `/api/v1/insights`, `/api/v1/product-details`) enabling multi-turn conversational category RAG with dynamic chip disambiguation powering the MSC IQ assistant.
- **Multimodal Text & Image Insight Pipeline (Amazon & ONDC):** Designed foundational category intelligence combining quantitative statistical analysis with Gemini 2.5 Flash inference for Amazon and ONDC product listings.

Projects

Privacy-Preserving Federated Sequential Recommendation (M.Tech Dissertation) — August 2025 – May 2026

- **Tools:** Python, PyTorch, Differential Privacy, Federated Learning, Contrastive Learning
- Researched and built a privacy-preserving sequential recommendation architecture preventing user trajectory leakage.
- Formulated **LDP-SeqMask** (Locally Differentially Private Sequence Masking), injecting calibrated noise into interaction sequences locally so raw client item sequences are never transmitted to central servers.
- Replaced unstable Exponential Mechanism item selection with a **calibrated Laplace perturbation mechanism**, eliminating gradient divergence and stabilizing federated model aggregation.

- Integrated **View-based Contrastive Learning (View-CL)** to maintain high recommendation accuracy (NDCG@10) even under heavy differential privacy noise, alongside sequence-aware heterogeneous client model selection.

Hermes IDS: Hybrid 2D-CNN & Random Forest Intrusion Detection System May 2025 – June 2025

- **Tools:** Python, PyTorch, Scikit-learn, Tab2Image, Tkinter, Pandas
- Built an autonomous AI threat detection and elimination engine converting tabular NetFlow records into 2D grayscale feature representations (Tab2Image) for spatial feature extraction via CNN.
- Trained and benchmarked CNN and ensemble models on the **UGRansome 2024** and **UNSW-NB15** datasets, achieving **98.6% detection accuracy**.
- Developed an interactive Tkinter SOC console for real-time packet inspection, anomaly visualization, and automated mitigation.

Intrusion Detection System via Heuristic Genetic Algorithm Feature Selection March 2025 – April 2025

- **Tools:** Python, Scikit-learn, XGBoost, Genetic Algorithms (GA), NumPy, Pandas
- Engineered an intrusion detection model on the **Bot-IoT** dataset utilizing Genetic Algorithm (GA) heuristic feature optimization.
- Eliminated redundant network features by over 70% while preserving $>$**99% attack detection recall** using XGBoost.

Cartoonify Image March 2022 – April 2022

- **Tools:** Python, OpenCV, NumPy, Tkinter
- Implemented image cartoonification using OpenCV and K-means clustering.
- Used elbow method to determine cluster count and combined bilateral filtering with Canny edge detection.

Indian Election Analysis October 2021 – November 2021

- **Tools:** Python, Pandas, Tweepy, TextBlob, Scikit-learn
- Built an NLP and ML workflow for Indian election analysis using live Twitter API data.
- Applied TextBlob polarity scoring and different ML models for positive/negative tweet classification.

Technical Skills

Generative AI & LLMs: LangGraph, Multi-Agent Supervisors, Gemini 2.5 Flash (Text & Vision), OpenAI Embeddings (`text-embedding-3-large`), RAG Systems, Prompt Engineering, Multimodal AI, Agentic Workflows

Machine Learning & Deep Learning: PyTorch, Scikit-learn, XGBoost, CNNs, Tab2Image, Contrastive Learning, Federated Learning, Locally Differential Privacy (LDP)

Backend & Data Systems: Python (FastAPI, Pydantic, Asyncio), PostgreSQL, `pgvector`, MongoDB Atlas, SQL, REST APIs

Data Engineering & Scraping: Zyte API (1,000 req/min), Anti-Bot Rotation, Threaded Ingestion, Scheduled Linux Cron Schedulers, Pandas, NumPy

Languages & Tools: Python, C++, SQL, Git, GitHub, Linux/Bash, VS Code, Postman, Splunk, Wireshark

Certifications

Google Cybersecurity Specialization (Coursera)	December 2024 – March 2025
Certified Python Developer Associate (PDAC)	March 2024
Data Science for Engineers (NPTEL)	January 2022 – March 2022
Machine Learning (Teachnook)	February 2022 – March 2022